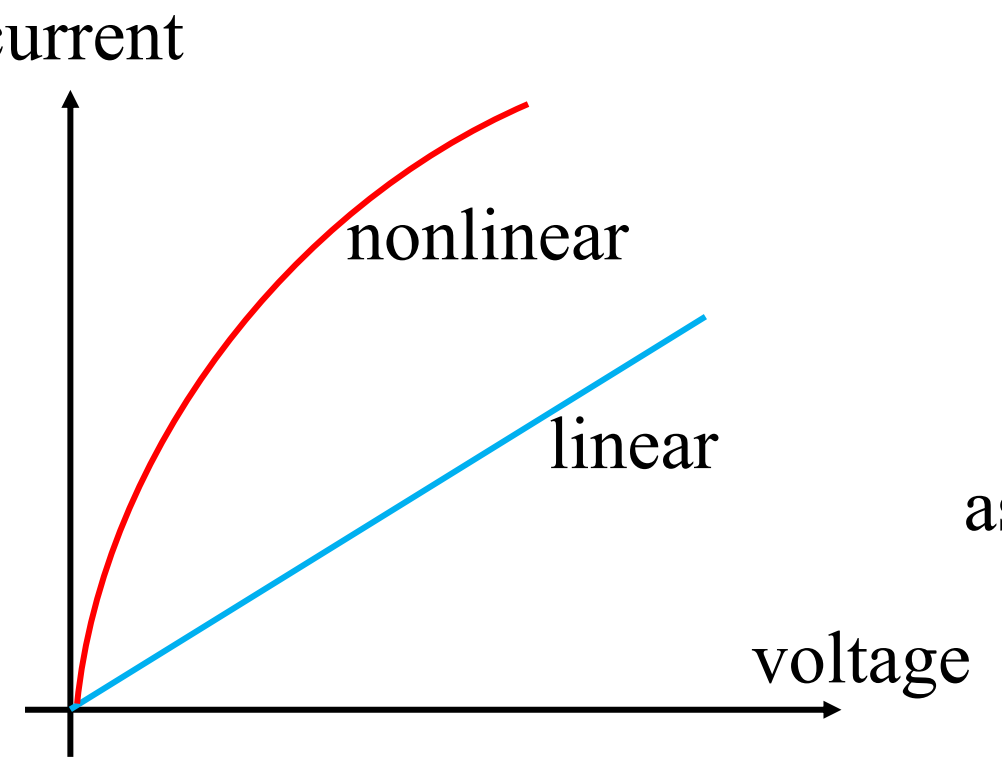
EEME 30004
Communications & Networks

Nonlinearity

Dr Sean Gu

Nonlinearity

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linear vs nonlinear

output input

resistor: $I = V/R$ inductor, capacitor

polynomial expression:

$$I = a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

assuming input V as a continuous wave (CW):

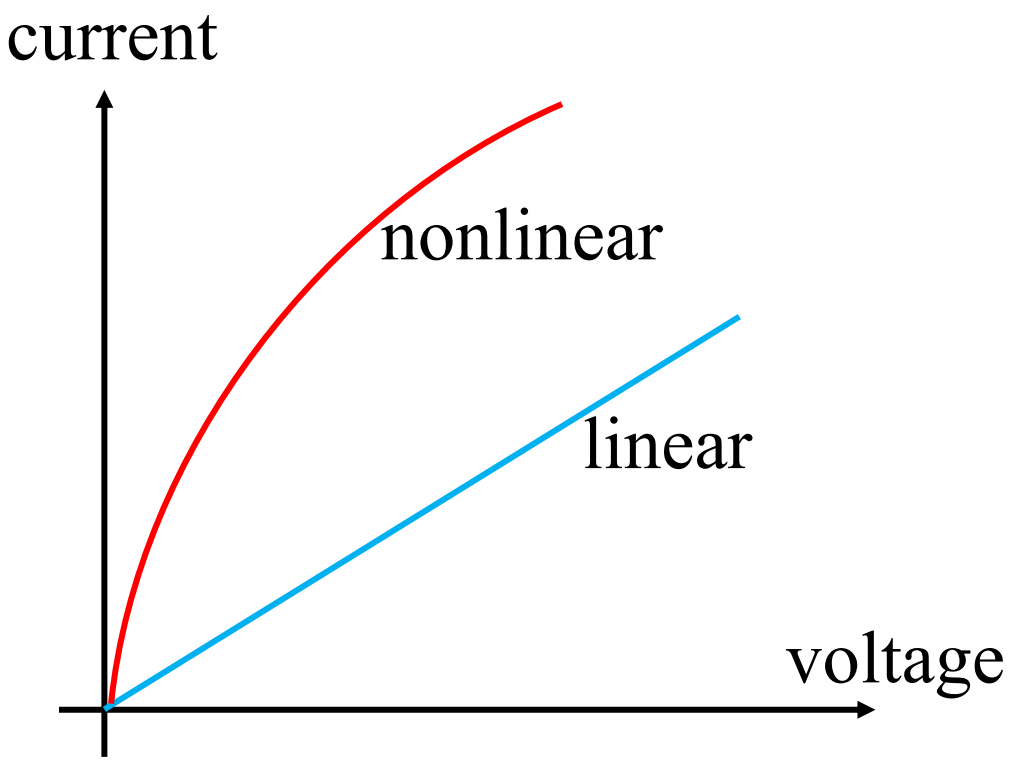
$$V(t) = V_m \cos(2\pi f \cdot t)$$

linear: $I(t) = V_m \cos(2\pi f \cdot t)\,/R$

what about the nonlinear case?

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linear vs nonlinear

polynomial expression:

$$I = a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

$$I(t) = a_2 \cdot V_m^2 \cos^2(2\pi f \cdot t)$$

$$= a_2 \cdot V_m^2 \frac{1 + \cos(2\pi 2f \cdot t)}{2}$$

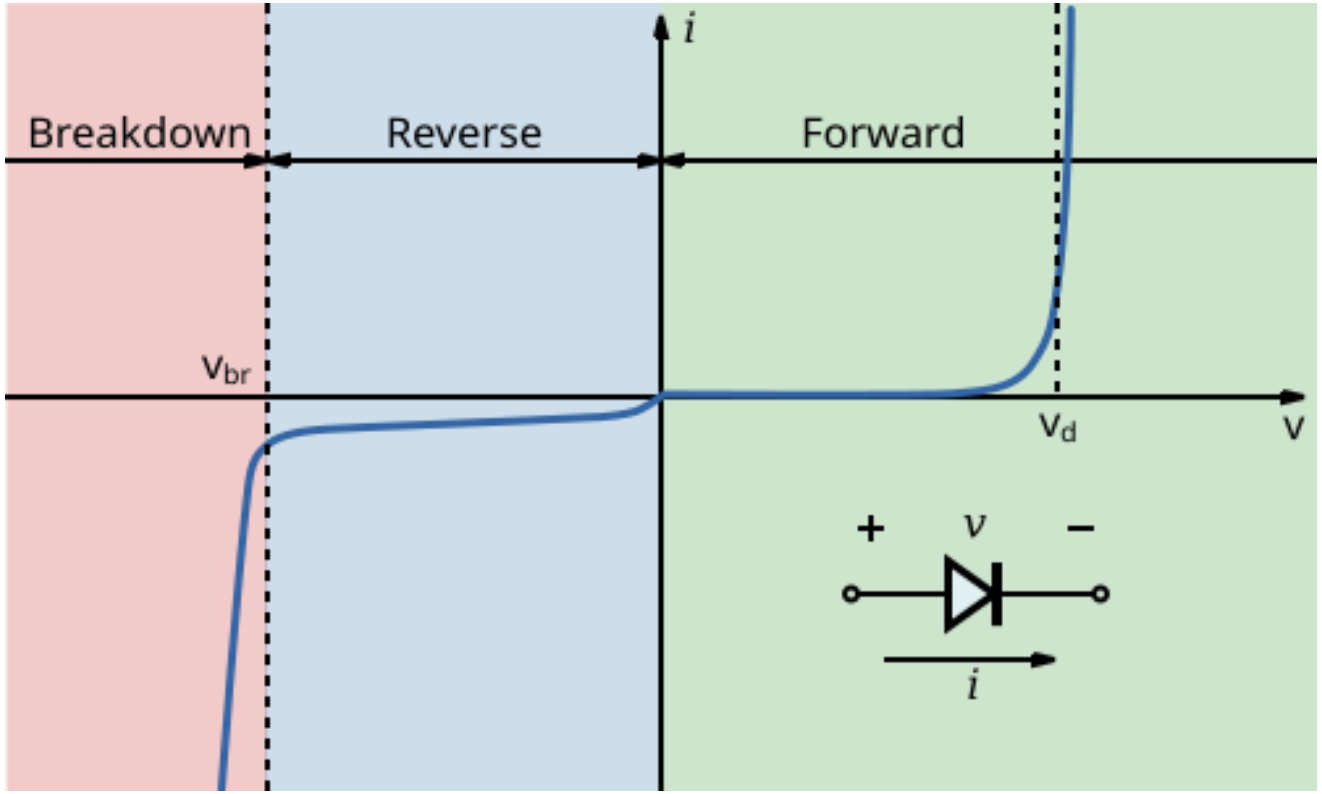
$$= \underbrace{\frac{1}{2} a_2 \cdot V_m^2}_{\text{DC}} + \underbrace{a_2 \cdot V_m^2 \frac{\cos(2\pi 2f \cdot t)}{2}}_{\text{second harmonic}}$$

linear: $I(t) = V_m \cos(2\pi f \cdot t)\,/R$

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diode current (I) – voltage (V) curve

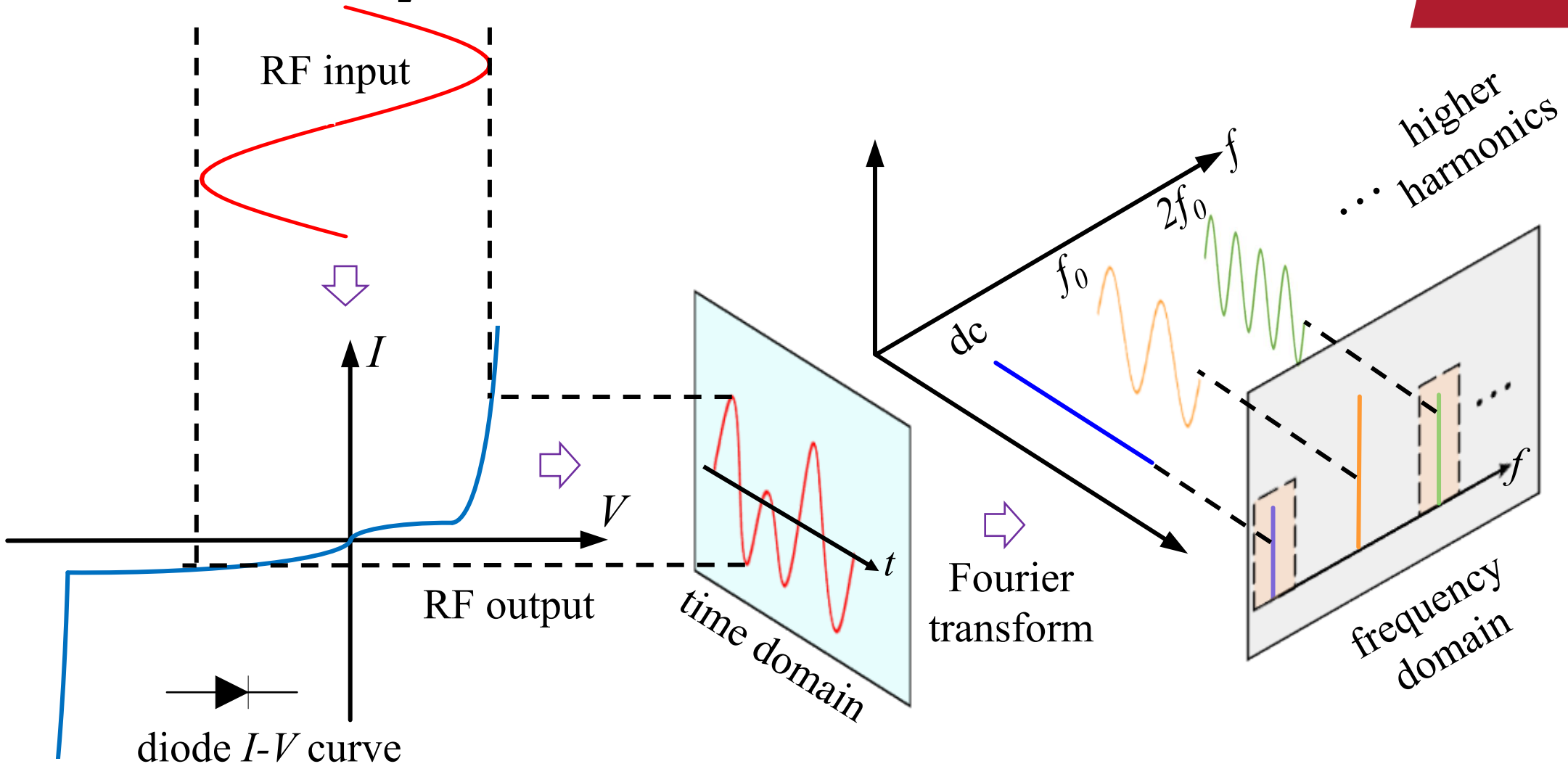




semiconductor analyser

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p-n diode *I-V* curve expression: $i_D = I_S \cdot \left(e^{\left(\frac{v_D}{n \cdot V_T}\right)} - 1\right)$

v_D voltage across the diode, i_D diode current

I_S saturation current, n ideality factor, V_T thermal voltage

small signal model (p-n junction):

$$i_D = I_D + i_d$$

$$v_D(t) = V_D + v_d(t) = V_D + V_d \cos(\omega t + \varphi)$$

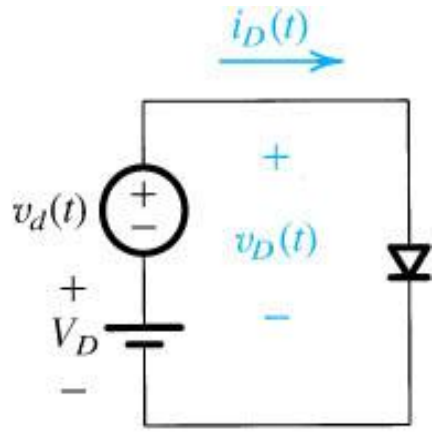
DC bias

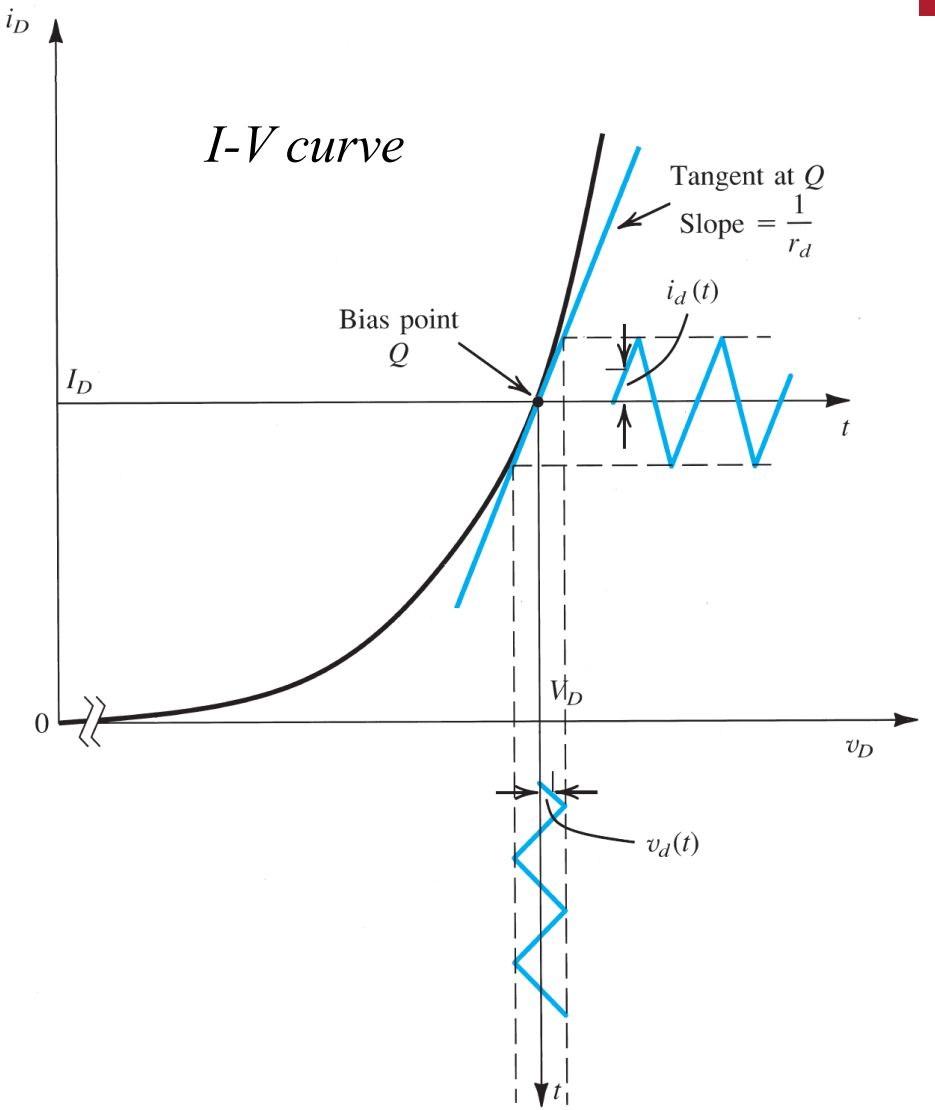
RF signals

$$i_D = I_S \cdot \left(e^{\left(\frac{V_D + v_d}{n \cdot V_T}\right)} - 1\right)$$

$$i_D = I_S \cdot e^{\left(\frac{V_D}{n \cdot V_T}\right)} \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S$$

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S$$





Nonlinearity

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)} - I_S \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)}$$

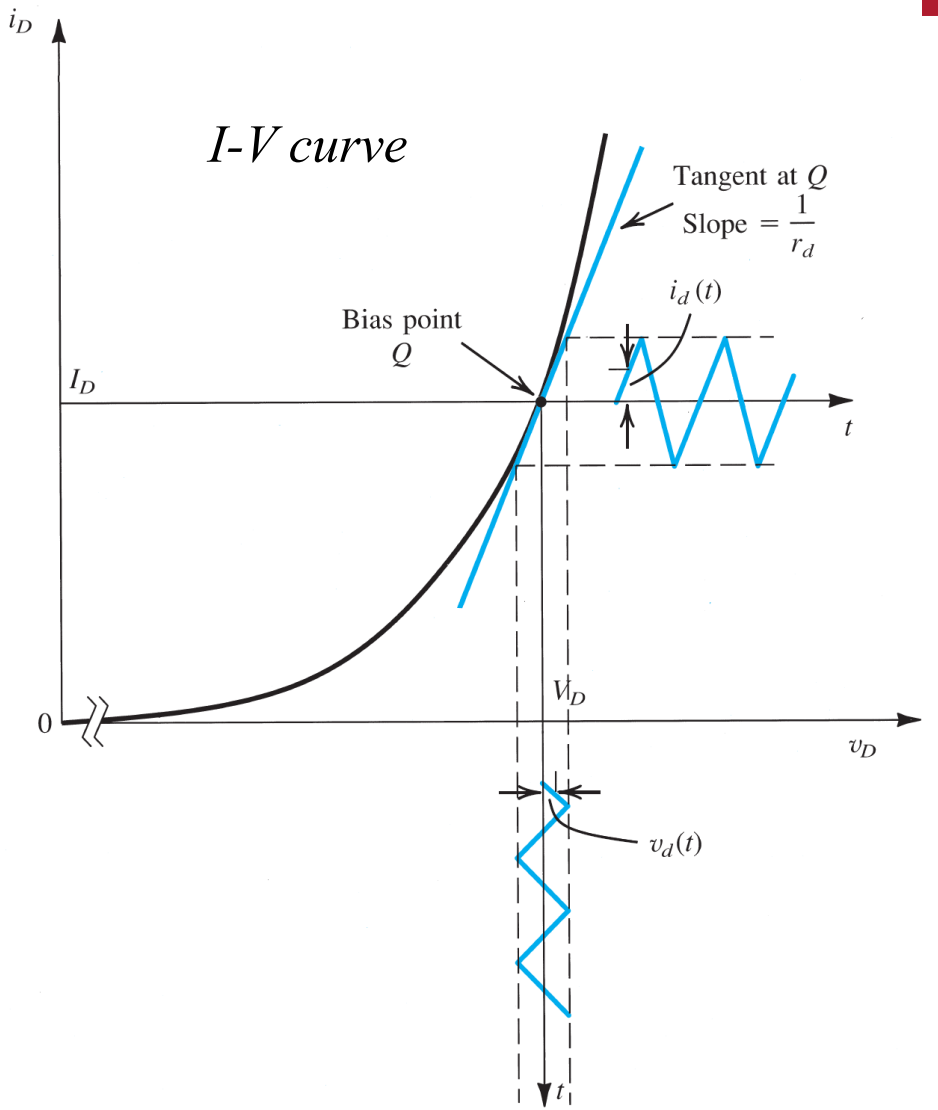
Taylor's serial expansion of $e^{\left(\frac{v_d}{n \cdot V_T}\right)}$: $\quad (e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!})$

$$i_D \approx I_D \cdot e^{\left(\frac{v_d}{n \cdot V_T}\right)}$$

$$\approx I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right) + \left(\frac{v_d}{n \cdot V_T}\right)^2 /2! + \left(\frac{v_d}{n \cdot V_T}\right)^3 /3!\right) + \cdots$$

$$i_D \approx I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right)\right) \quad \text{if} \quad v_d \ll nV_T$$

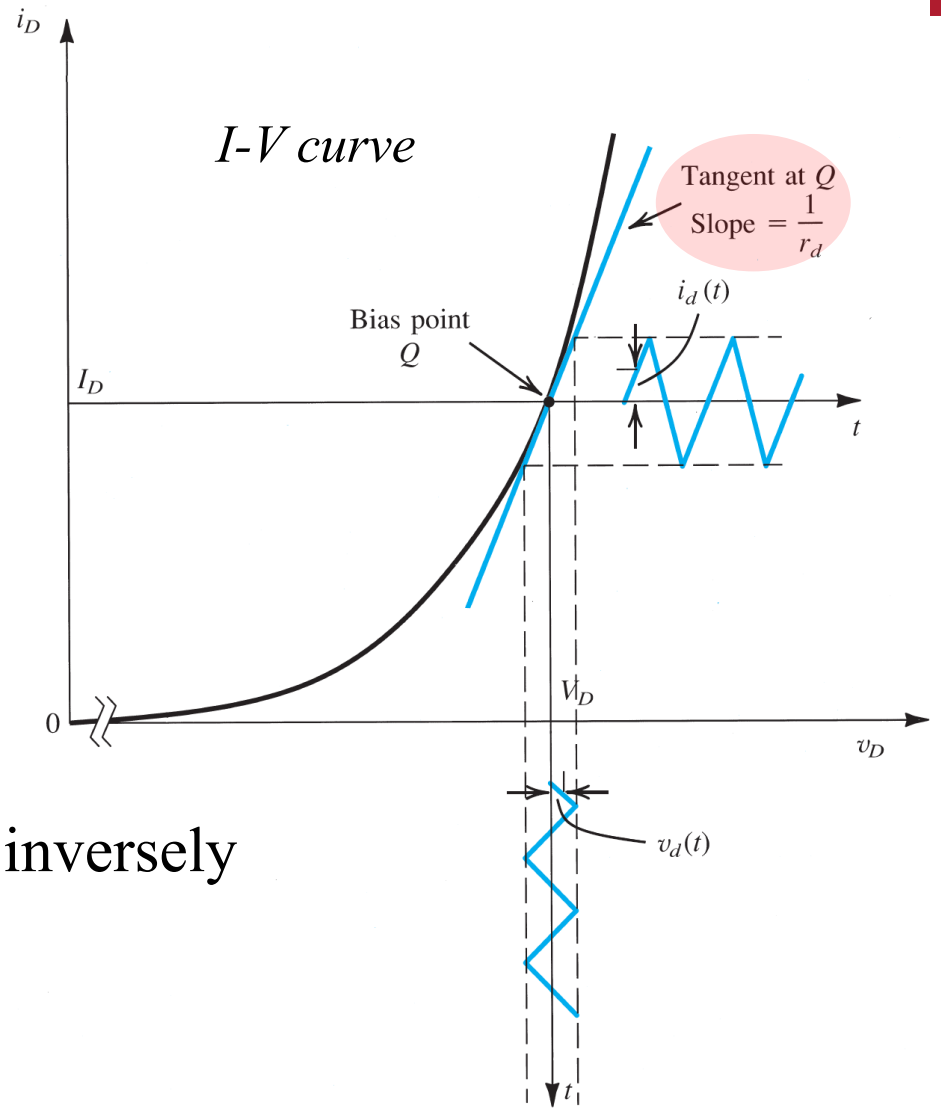
We consider that this condition is satisfied if $v_d \leq 1$



Nonlinearity

since $i_D = I_D + i_d = I_D \cdot \left(1 + \left(\frac{v_d}{n \cdot V_T}\right)\right)$

we have $i_d = I_D \cdot \left(\frac{v_d}{n \cdot V_T}\right)$



- Dynamic resistance of the diode by definition:

$$r_d = {}^{v_d}/_{i_d} = {}^{n \cdot V_T}/_{I_D}$$

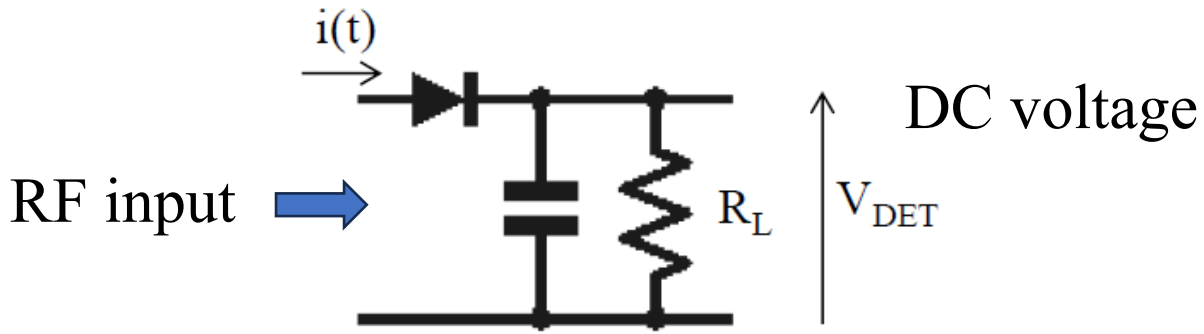
- Diode's small signal model: dynamic resistance r_d is inversely proportional to the quiescent current I_D.

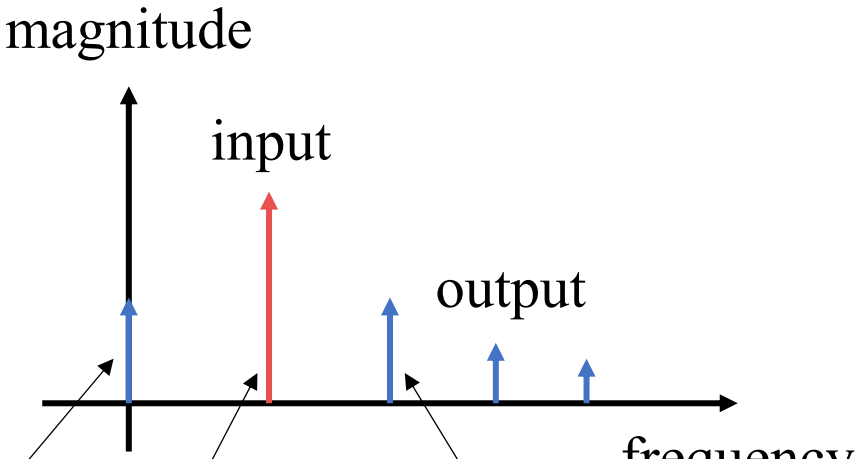
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applications in power detectors or energy harvesters:

our goal is to maximize the output of the DC component:





$$I_D = I_S \cdot \left(e^{\left(\frac{V_D}{n \cdot V_T}\right)} - 1 \right)$$

$$I_D = I_S \cdot \left(\frac{V_1}{n \cdot V_T} \cos(\omega t) + \frac{1}{2} \left(\frac{V_1}{n \cdot V_T} \right)^2 \cos^2(\omega t) + \cdots \right)$$

(zero bias or no bias)

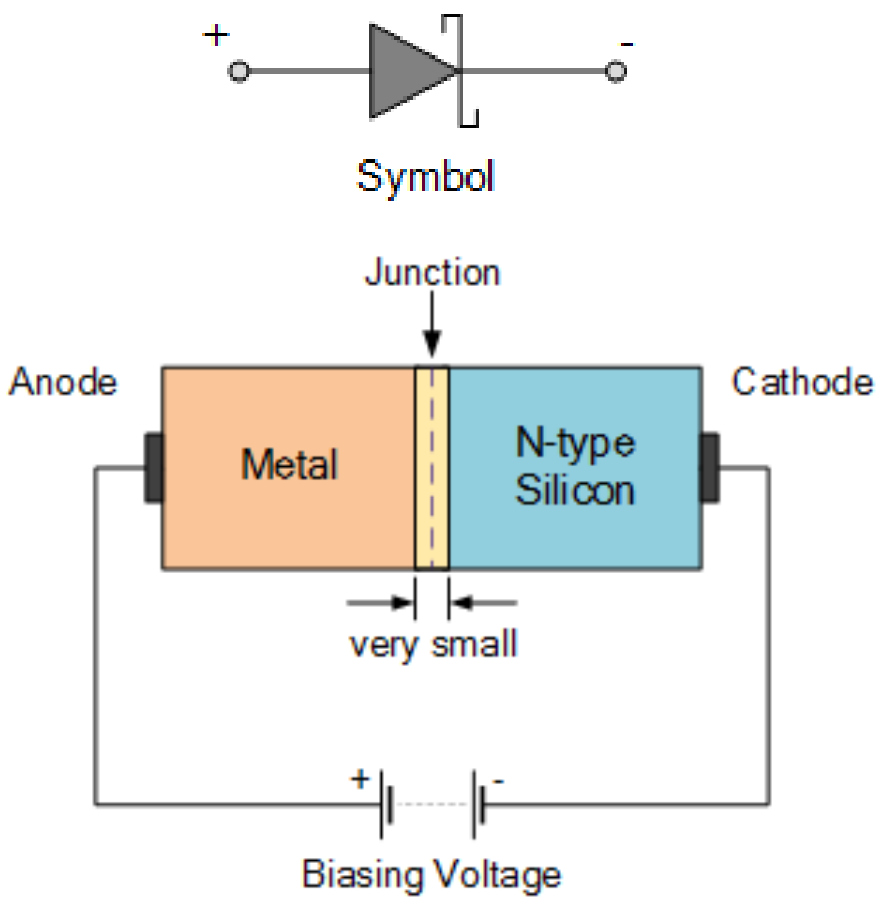
assume $V_D = V_1 \cos(\omega t)$

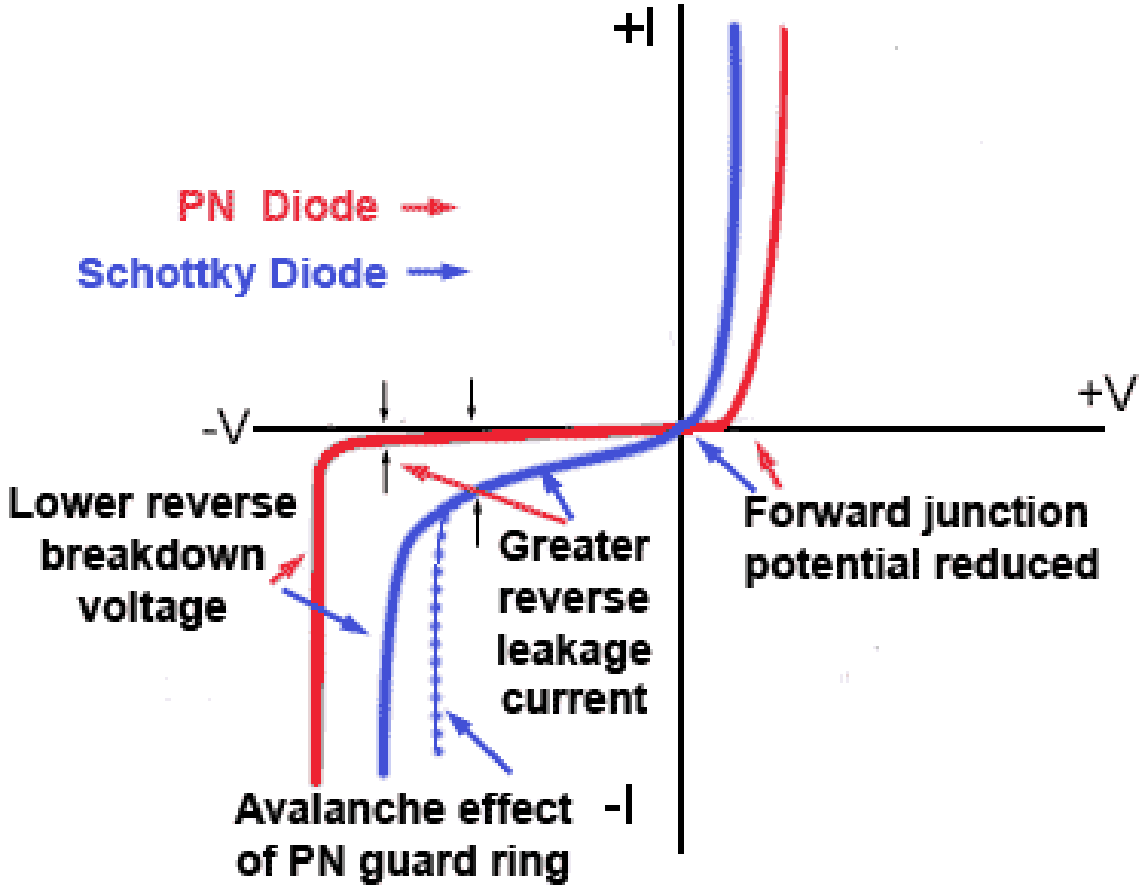
$$I_D \approx \frac{I_S}{4} \cdot \left(\frac{V_1}{n \cdot V_T} \right)^2 + I_S \frac{V_1}{n \cdot V_T} \cos(\omega t) + \frac{I_S}{4} \cdot \left(\frac{V_1}{n \cdot V_T} \right)^2 \cos(2\omega t)$$

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Schottky diode: no depletion layer, small knee voltage (0.2 to 0.5 V)

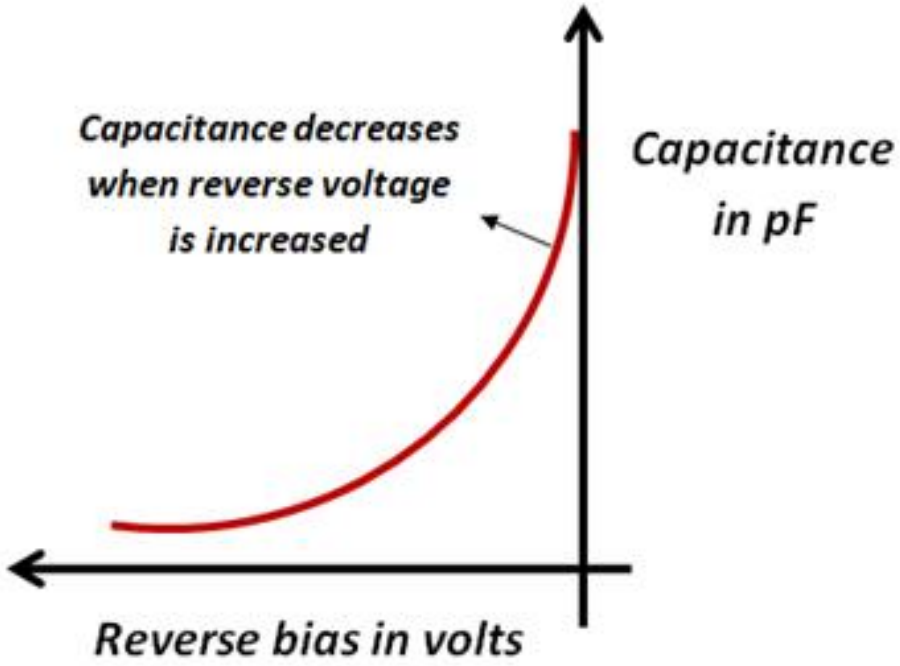




Nonlinearity

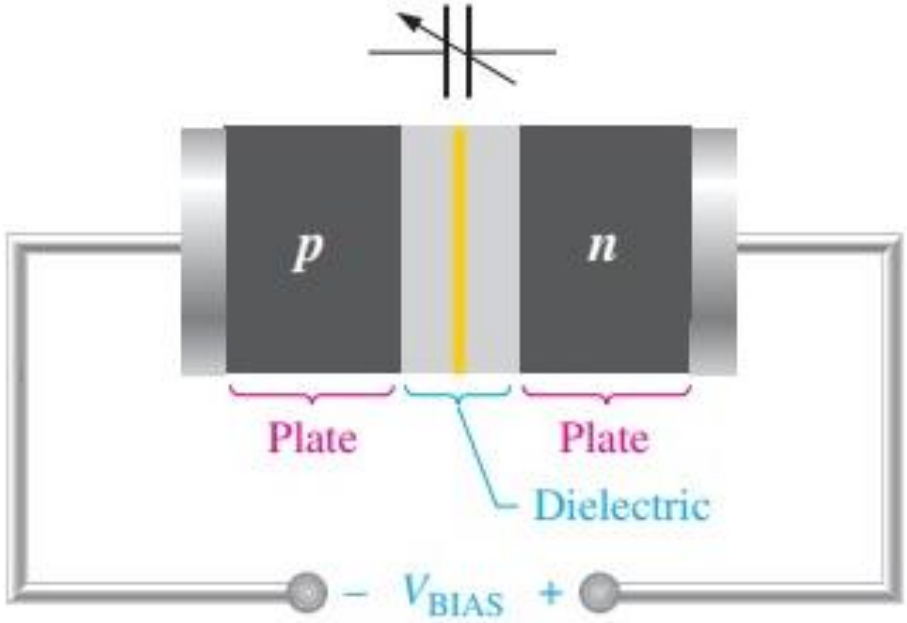
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varactor diode: voltage-controlled capacitance





Symbol of Varactor Diode



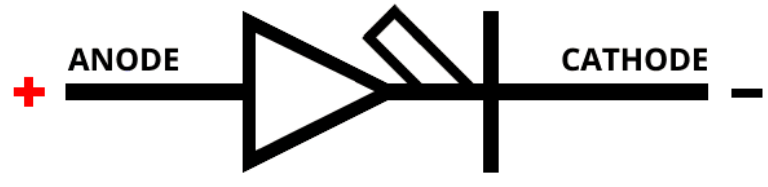
When varactor diode is **reverse biased**, its capacitance decreases as the reverse voltage increases. Hence, we have a variable capacitance as a function of the applied voltage.

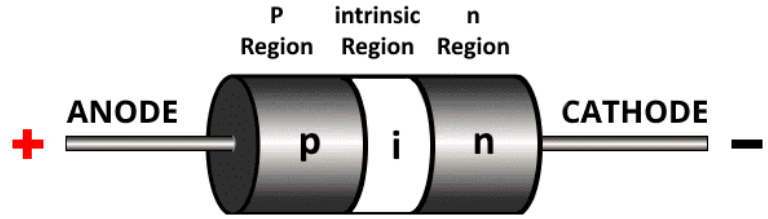
Nonlinearity

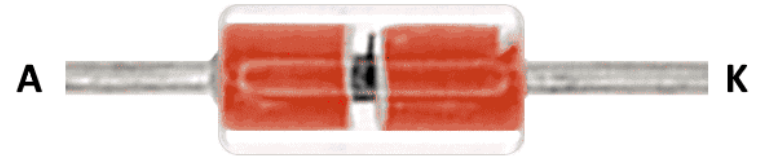
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PIN diode: RF switch

a diode with a wide, undoped intrinsic semiconductor region between a p-type semiconductor and an n-type semiconductor region. The p-type and n-type regions are typically heavily doped because they are used for ohmic contacts.



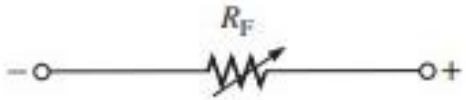




RF off



Reverse-biased PIN Diode



Forward-biased PIN Diode

RF on

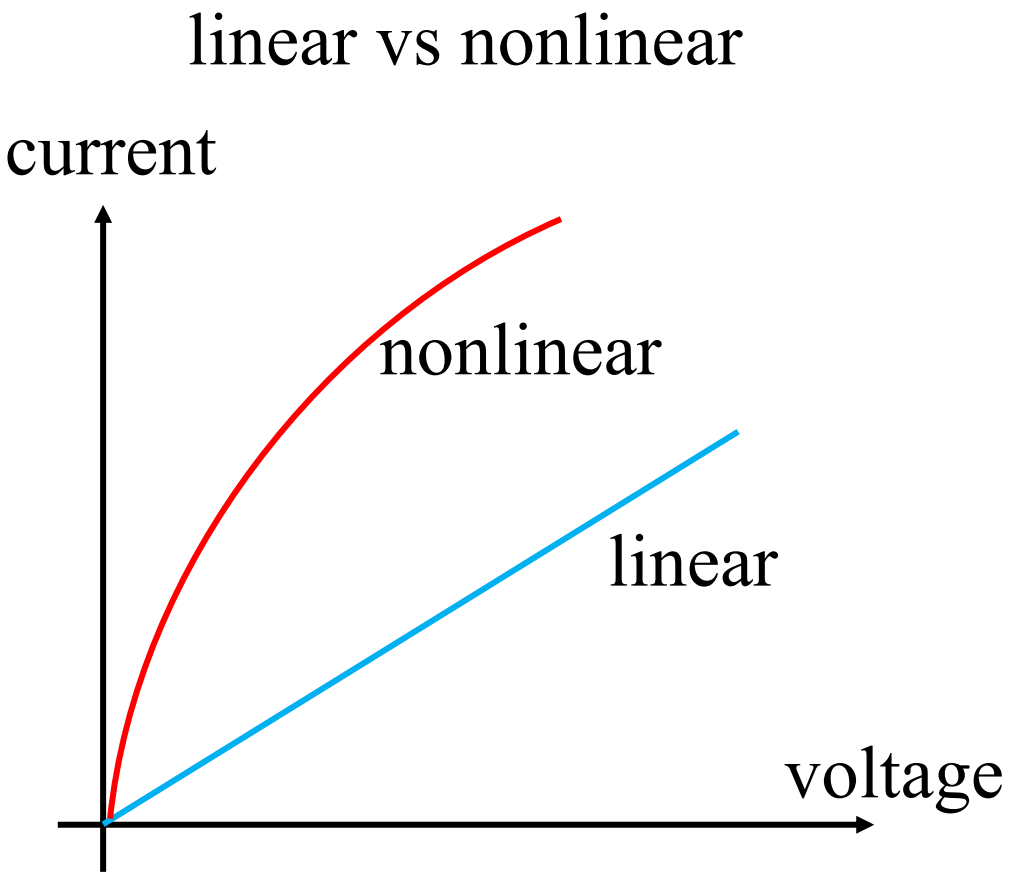
applications :
PIN diodes are used for magnitude and phase control of microwave signals.

advantages :

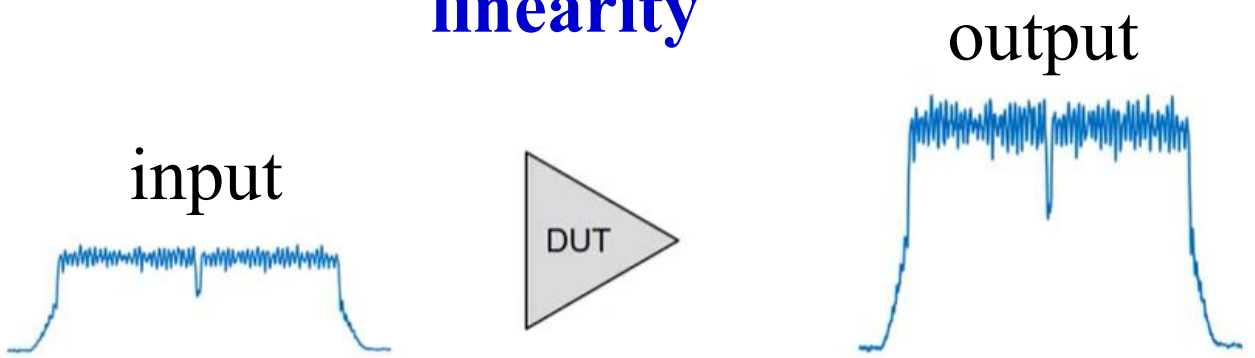
- good power handling capability & small power consumption
- fast switching & reliable

Nonlinearity

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linearity

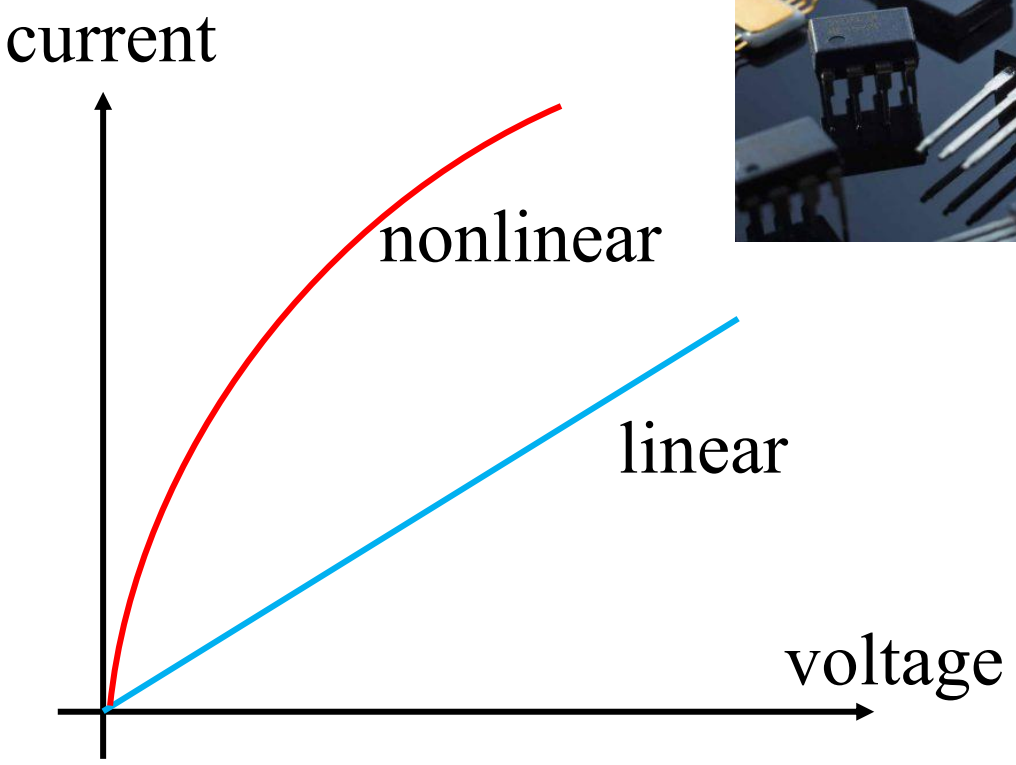


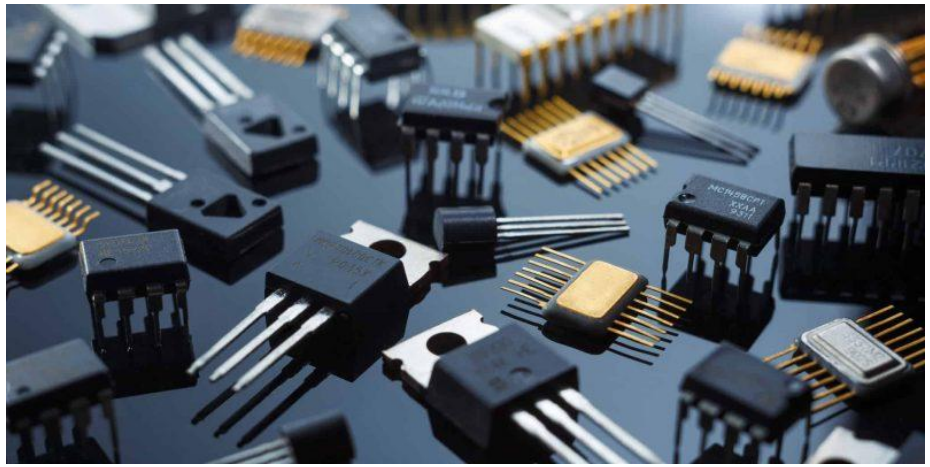
nonlinearity

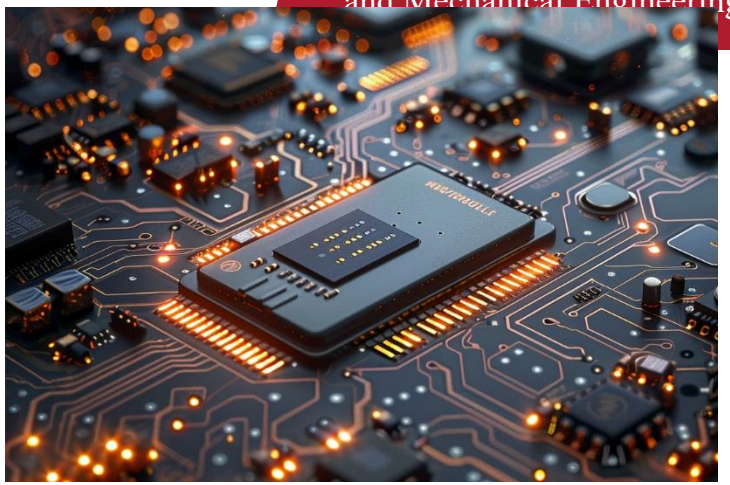
- Of great utility for amplification, detection, and frequency conversion.
- Leads to undesirable effects, including losses, signal distortion, and possible interference.

Nonlinearity

linear vs nonlinear







possible effects of nonlinearity:

- harmonic generation (2nd, 3rd, 4th, …)
- saturation (gain reduction in an amplifier)
- intermodulation distortion (products of a two-tone signal)
- spectral regrowth

Nonlinearity

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harmonic generation:

polynomial expression of the I – V relationship:

$$I = a_0 + a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

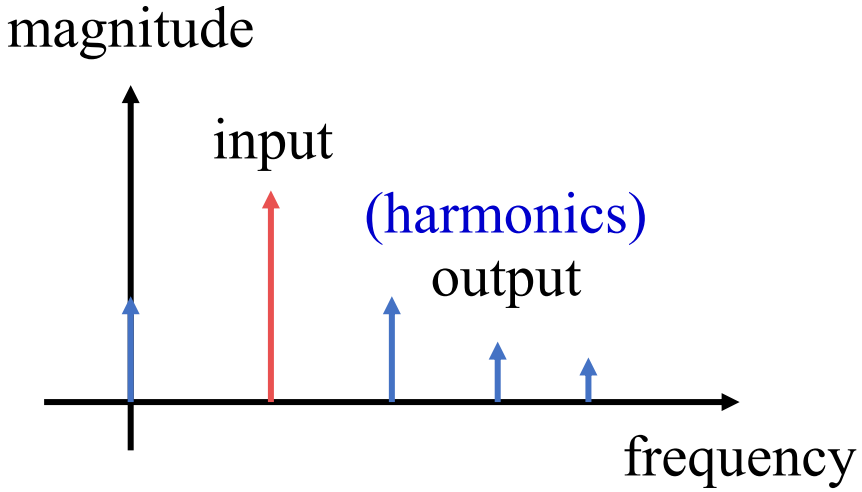
input signal (CW): $V_{in} = V_m \cos(2\pi f \cdot t)$

$$I = a_0 + a_1 \cdot V + a_2 \cdot V^2 + a_3 \cdot V^3 + a_4 \cdot V^4 + \cdots$$

$$I(t) = a_2 \cdot V_m^2 \cos^2(2\pi f \cdot t)$$

$$= a_2 \cdot V_m^2 \frac{1 + \cos(2\pi 2f \cdot t)}{2}$$

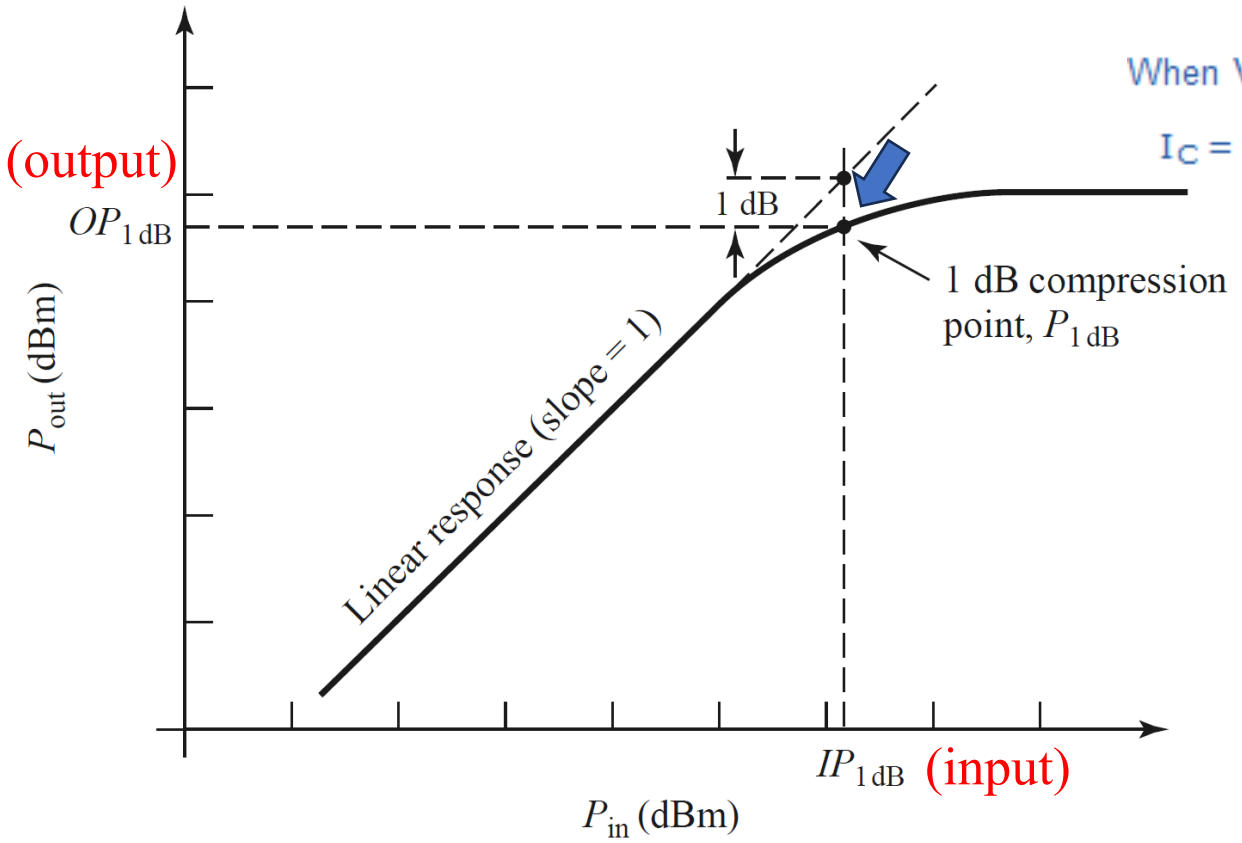
$$= \underbrace{\frac{1}{2} a_2 \cdot V_m^2}_{\text{DC}} + \underbrace{a_2 \cdot V_m^2 \frac{\cos(2\pi 2f \cdot t)}{2}}_{\text{second harmonic}}$$

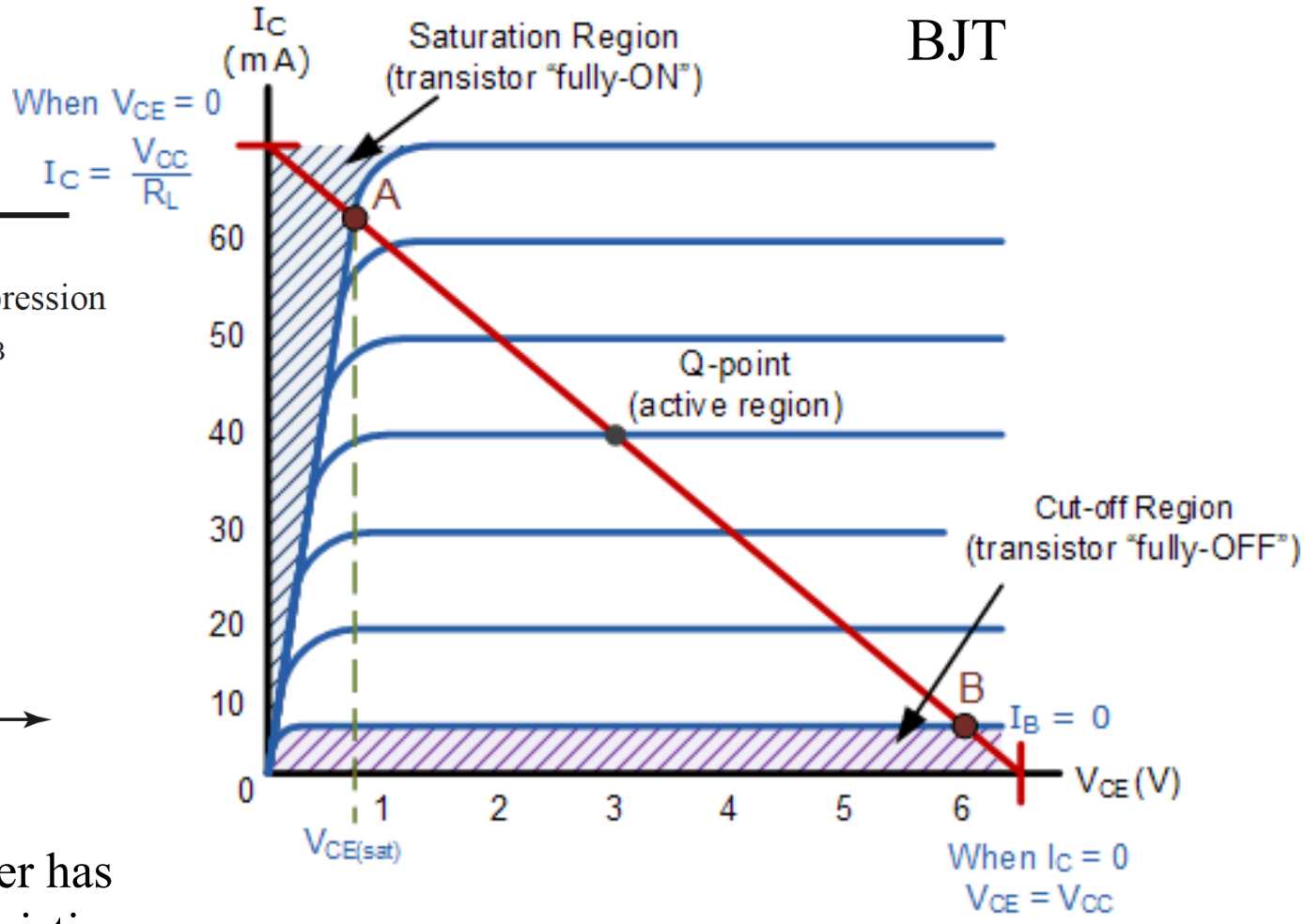


Nonlinearity

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saturation:



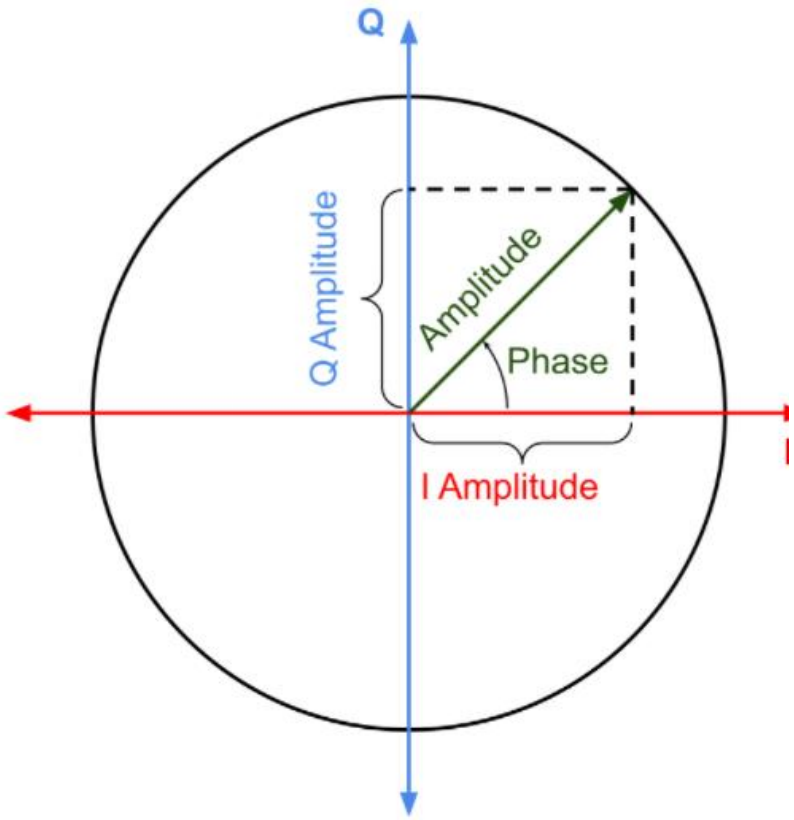


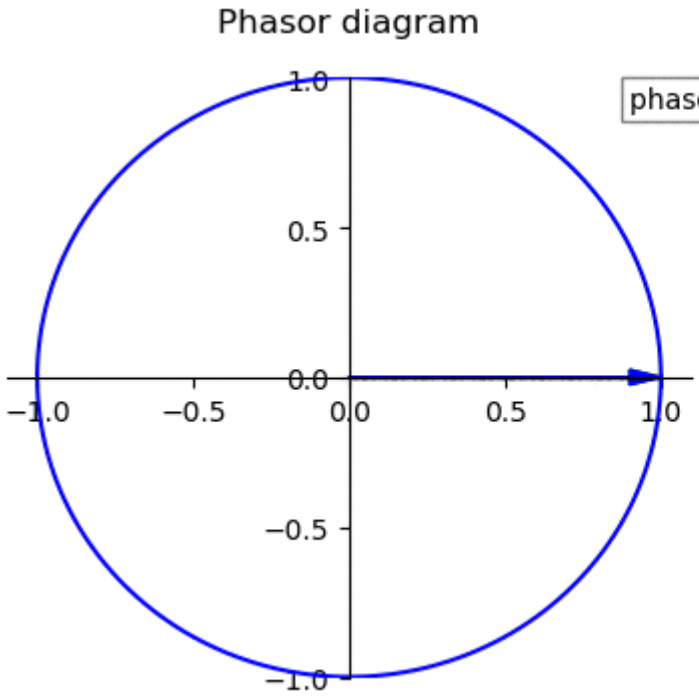
1 dB compression point: amplifier's output power has decreased by 1 dB from the ideal linear characteristic.

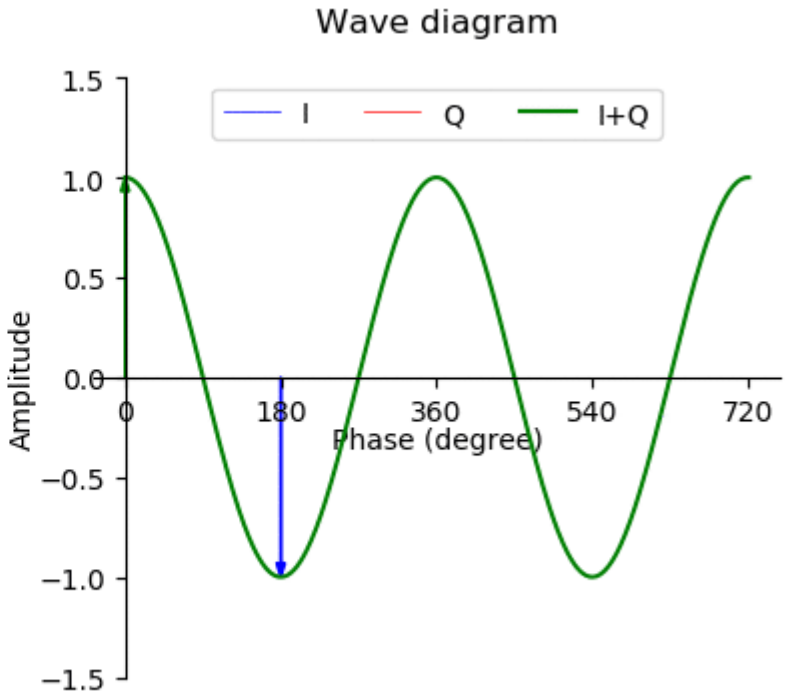
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distortion:





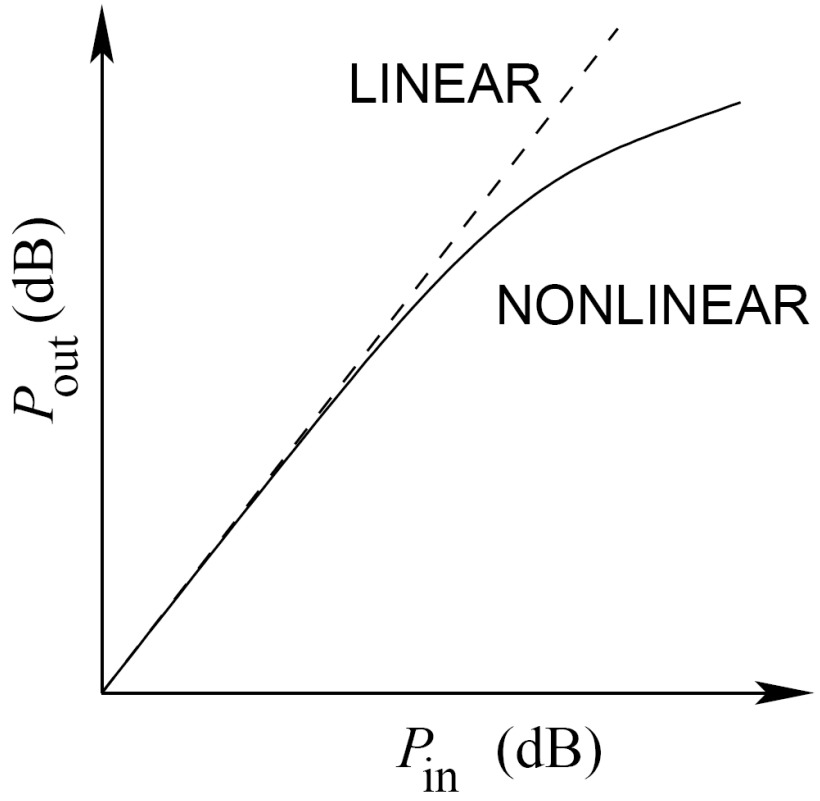


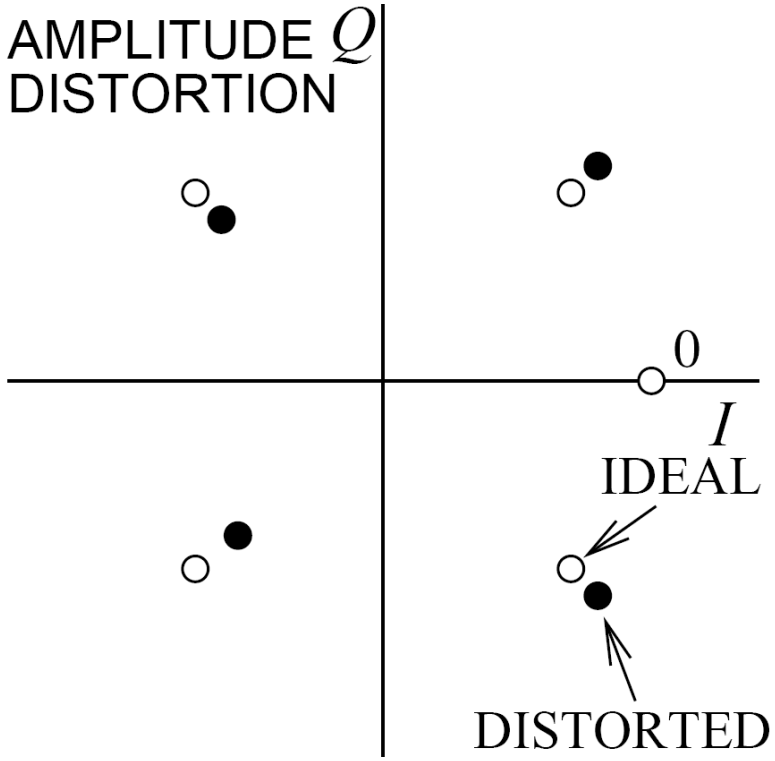
Efficient Representation of Complex Signals; Baseband Processing; Quadrature Amplitude Modulation (QAM); Reduced Hardware Complexity.

Nonlinearity

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amplitude distortion

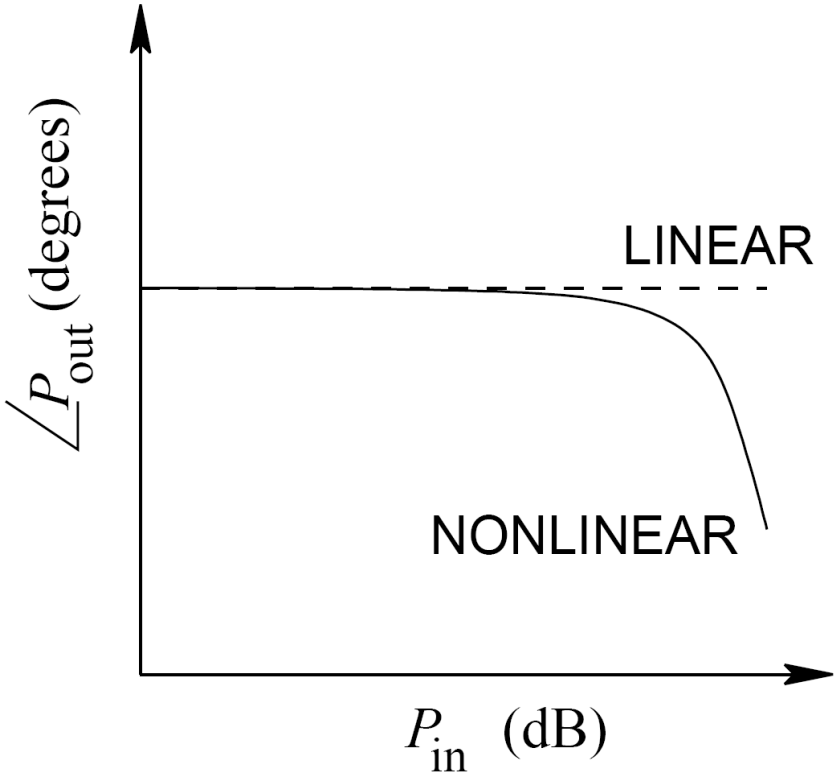


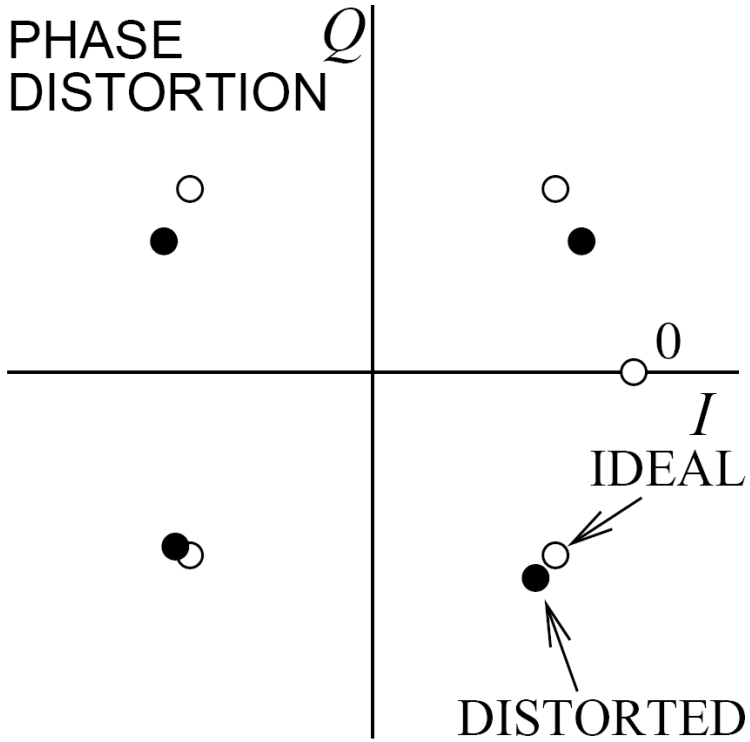


Nonlinearity

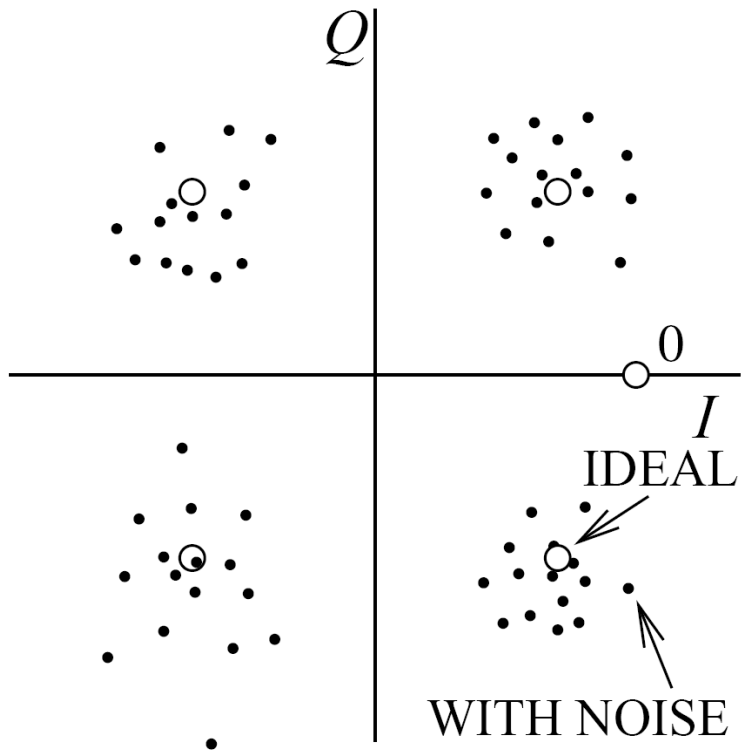
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phase distortion





noise distortion



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intermodulation

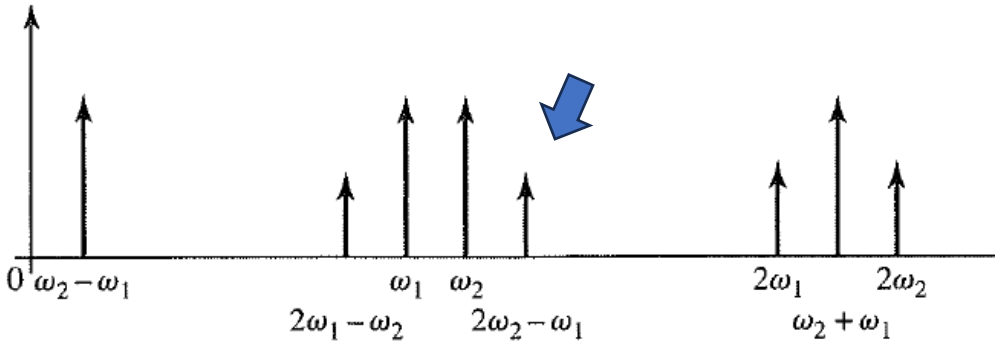
assume two CW inputs as:

$$v_i = V_0(\cos\omega_1 t + \cos\omega_2 t)$$

nonlinearity formulated by:

$$v_o = a_0 + a_1 \cdot v_i + a_2 \cdot v_i^2 + a_3 \cdot v_i^3 + a_4 \cdot v_i^4 + \cdots$$

$$\begin{aligned} v_o &= a_0 + a_1 V_0(\cos\omega_1 t + \cos\omega_2 t) + a_2 V_0^2(\cos\omega_1 t + \cos\omega_2 t)^2 + a_3 V_0^3(\cos\omega_1 t + \cos\omega_2 t)^3 + \cdots \\ &= a_0 + a_1 V_0 \cos\omega_1 t + a_1 V_0 \cos\omega_2 t + \tfrac{1}{2}a_2 V_0^2(1 + \cos 2\omega_1 t) \\ &\quad + \tfrac{1}{2}a_2 V_0^2(1 + \cos 2\omega_2 t) + a_2 V_0^2 \cos(\omega_1 - \omega_2)t + a_2 V_0^2 \cos(\omega_1 + \omega_2)t \\ &\quad + a_3 V_0^3\left(\tfrac{3}{4}\cos\omega_1 t + \tfrac{1}{4}\cos 3\omega_1 t\right) + a_3 V_0^3\left(\tfrac{3}{4}\cos\omega_2 t + \tfrac{1}{4}\cos 3\omega_2 t\right) \\ &\quad + a_3 V_0^3\left[\tfrac{3}{2}\cos\omega_2 t + \tfrac{3}{4}\cos(2\omega_1 - \omega_2)t + \tfrac{3}{4}\cos(2\omega_1 + \omega_2)t\right] \\ &\quad + a_3 V_0^3\left[\tfrac{3}{2}\cos\omega_1 t + \tfrac{3}{4}\cos(2\omega_2 - \omega_1)t + \tfrac{3}{4}\cos(2\omega_2 + \omega_1)t\right] + \cdots, \end{aligned}$$



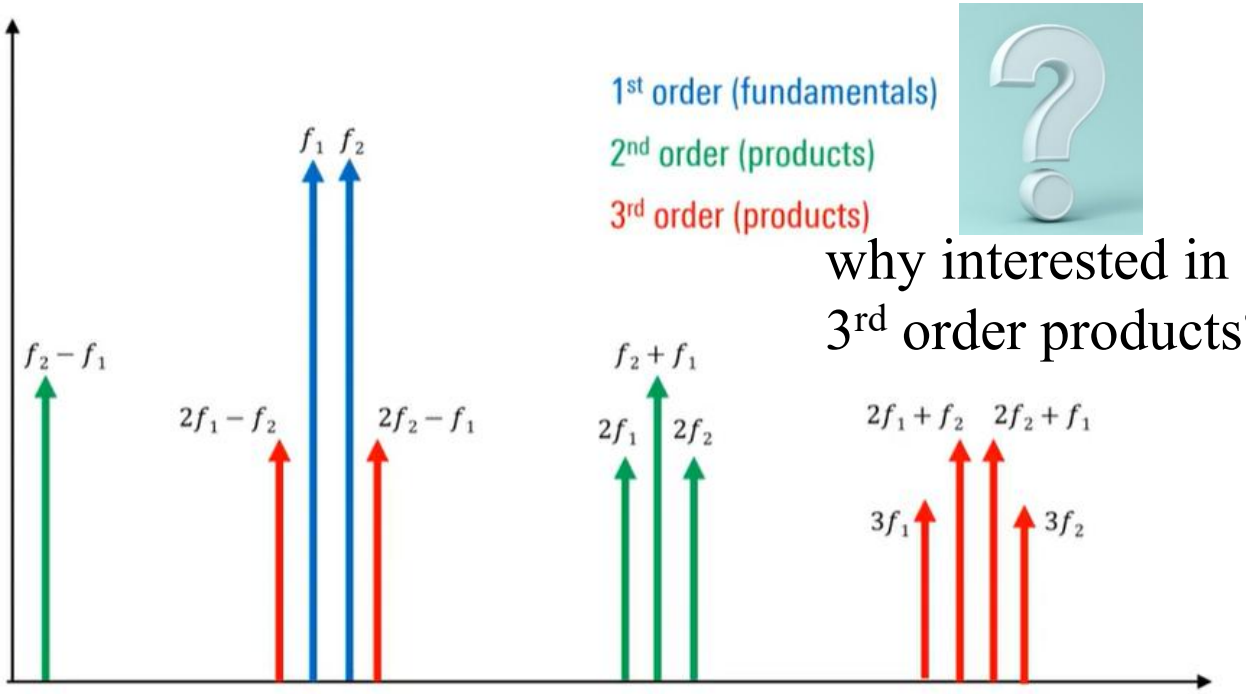
fundamental: ——— harmonics: ——— intermodulation: ———

Nonlinearity

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intermodulation

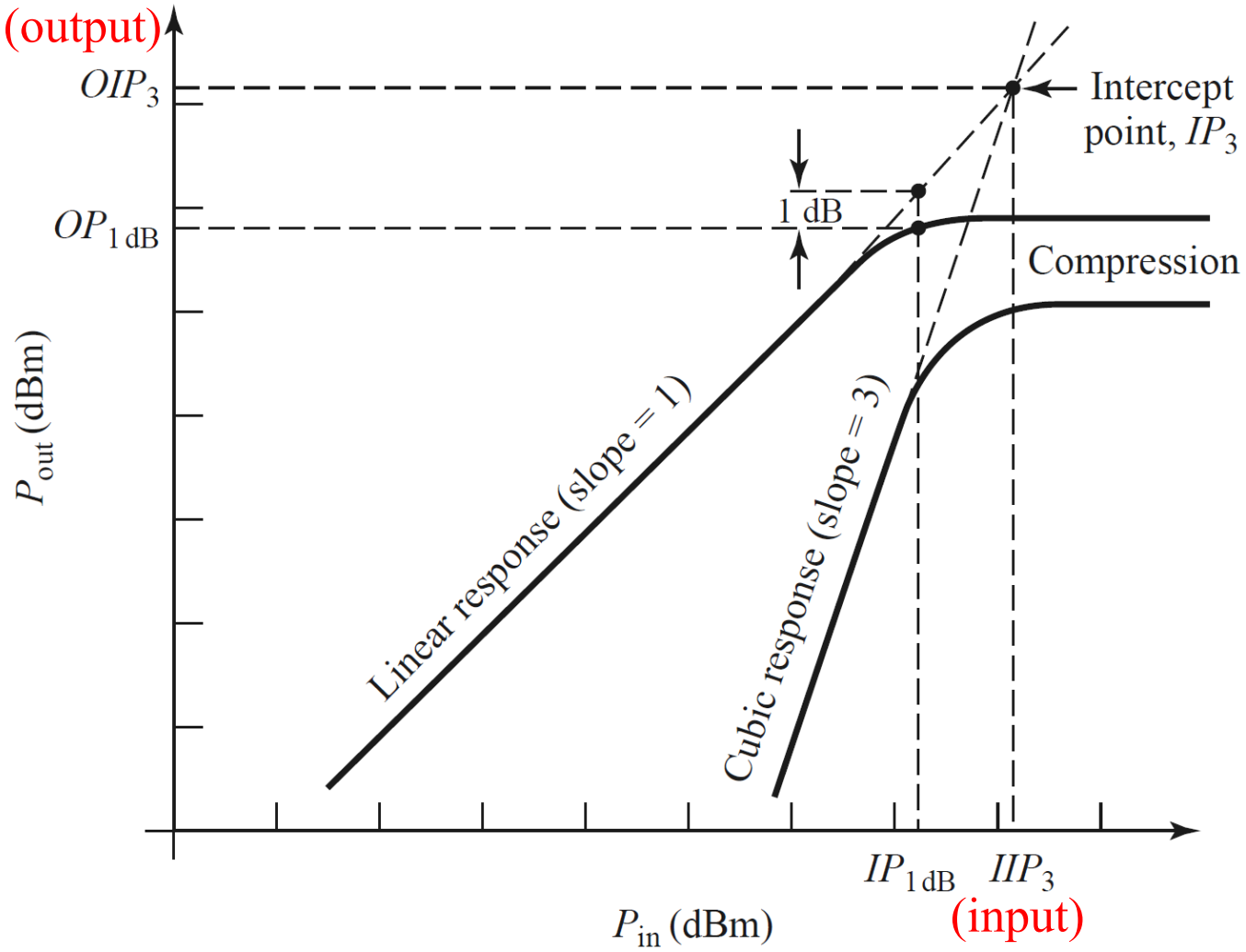
different orders of products in freq. domain



why interested in
3rd order products?

Nonlinearity

intermodulation



Third-order intercept point (IP3): a ***hypothetical*** point where the power of a device's fundamental signal and its third-order intermodulation (IM3) products would be equal if the device remained linear.

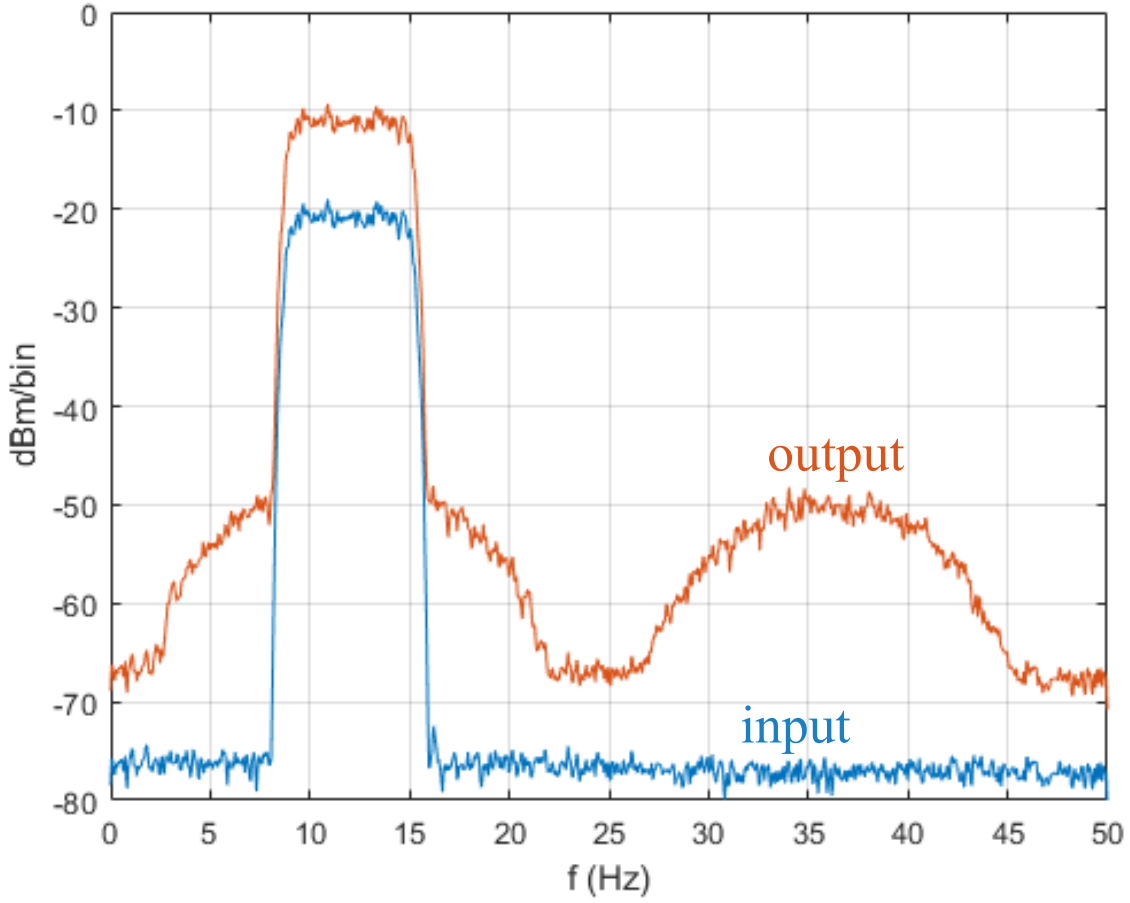
For each 1 dB increase in the fundamental signal, the third order (unwanted) intermodulation products increase in level by 3 dB.

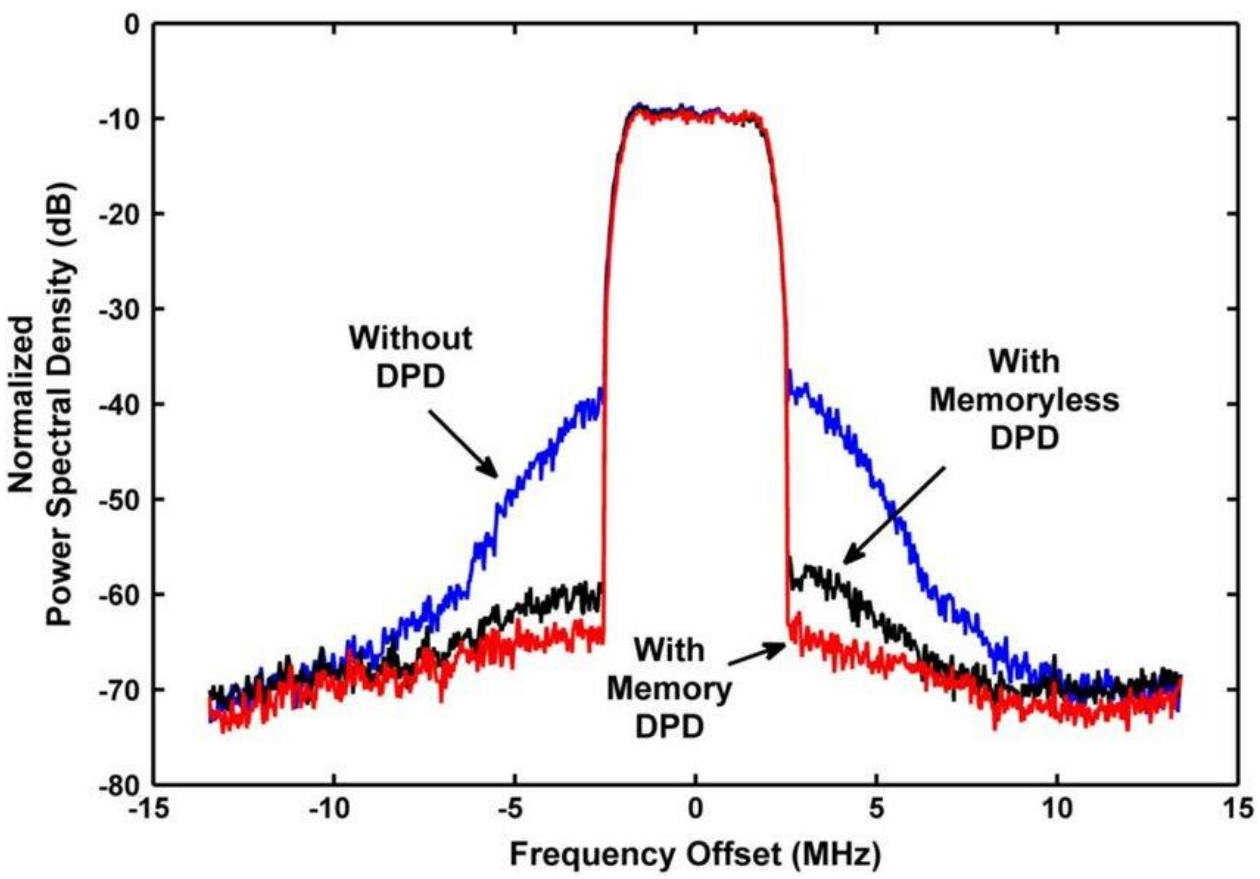
Second-order products and harmonics are not plotted in this graph as they are not in the passband of fundamental signals.

Nonlinearity

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spectral regrowth:





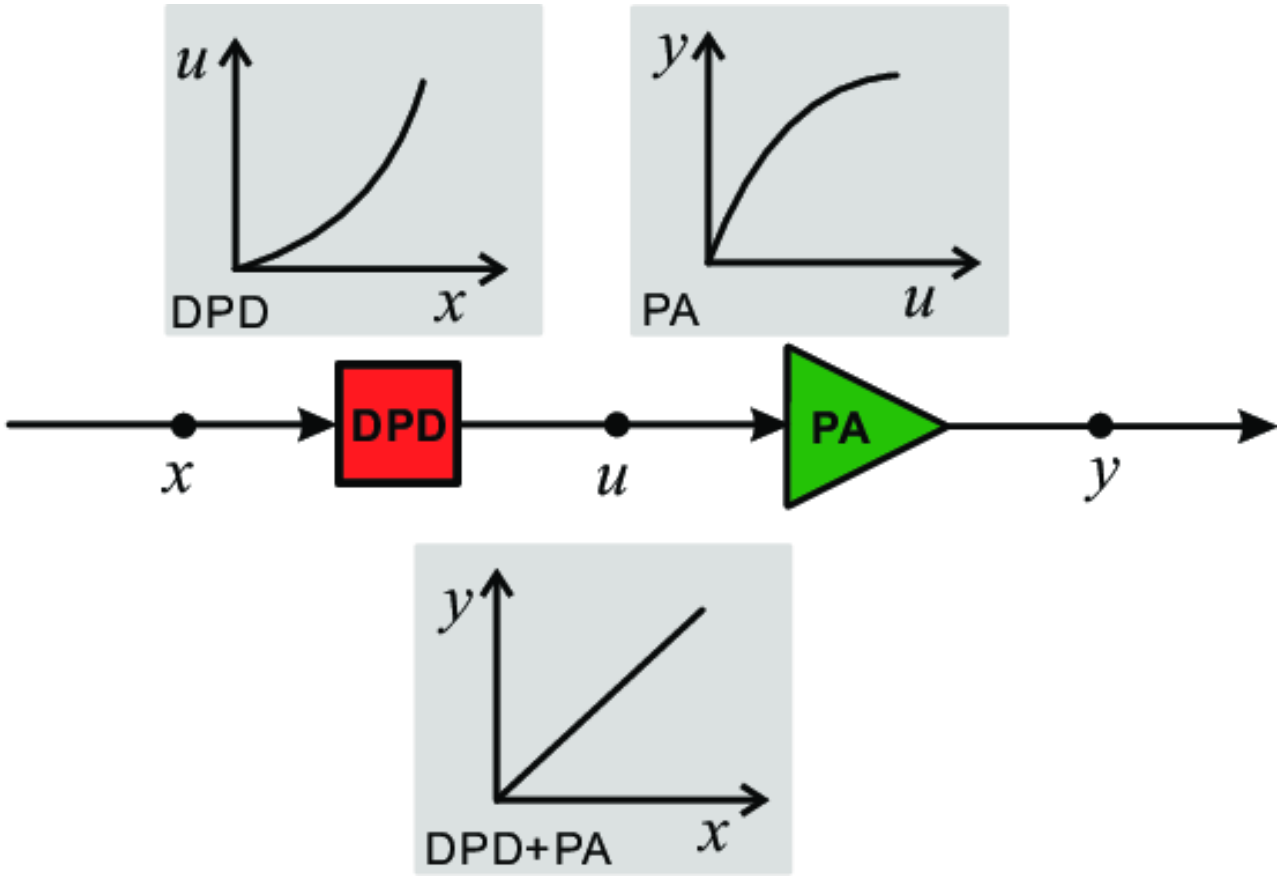
modeled QAM-16 spectra at input and output of amplifier with 3rd-order distortion

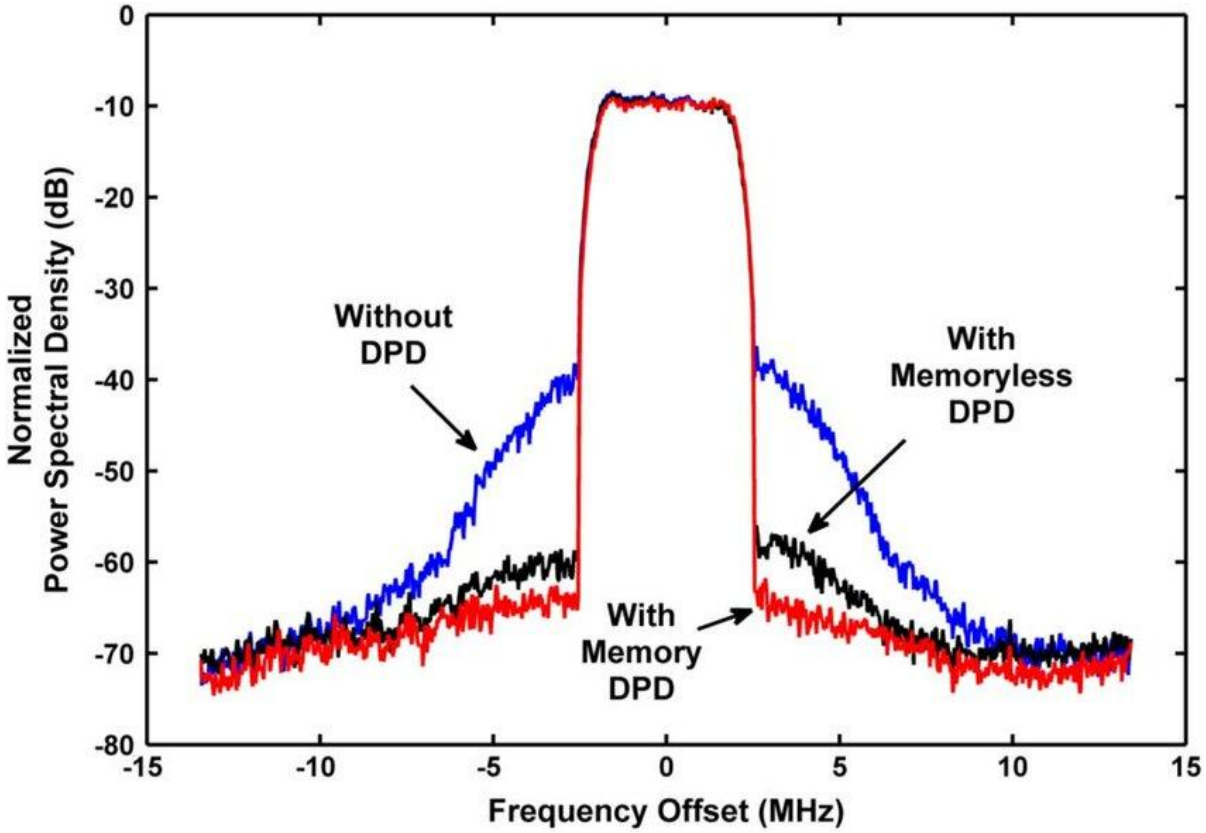
digital pre-distortion (DPD)

Nonlinearity

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digital pre-distortion (DPD)

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Questions?

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